

Abhinav Nandwani

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EDUCATION

University of Wisconsin-Madison

B.S Computer Engineering and Computer Science (Double Major)

GPA: 3.9/4.0

Madison, WI

Expected May 2027

EXPERIENCE

Advanced Micro Devices (AMD) — Applied AI Labs

Co-op

Boxborough, MA

August 2025 - Present

- Bootstrapped AMD's internal AI benchmarking function, building evaluation harnesses for agentic systems (Claude Code, GPT-5 Codex, Gemini, and internal agents) on silicon design tasks — spec-to-RTL, waveform debug, design verification — using verifiable rewards via UVM testbenches and formal equivalence checks; results directly inform agent improvement cycles.
- Built agentic waveform debug tool deployed to 1,000+ engineers across AMD's chip design teams, cutting debug cycles from 2 weeks to 2 days; developed waveform MCP enabling agents to trace signals, analyze waveforms, and debug simulation runs directly.
- Extended benchmarking expertise to evaluate external AI vendors and startups building tools for chip design, directly informing AMD leadership on adoption and acquisition decisions.

Professor Daifeng Wang Lab

Undergraduate Researcher

Madison, WI

January 2025 - August 2025

- Built inference and fine-tuning pipelines for single-cell genomics foundation models processing 4M+ profiles from PsychAD Consortium dataset (2.5M cells, 3 modalities), reducing adaptation time by 40%.
- Developed phenotype classification models achieving 90%+ accuracy with clustering and trajectory inference discovering 5+ novel cell-state transitions in Alzheimer's progression.
- Built distributed training infrastructure on CHTC GPU Lab across 10+ NVIDIA H200 GPUs with Docker/Singularity workflows, reducing deployment overhead by 60%.
- Optimized multi-GPU fine-tuning with PyTorch DDP achieving 3.2x faster convergence. Scaled PEFT (LoRA/QLoRA) from 4 to 32 GPUs, reducing costs 75%.

PROJECTS

Xilinx FPGA AI Inference Accelerator

- Architected hardware-accelerated LLM inference engine on AMD Xilinx ZYNQ 7020 FPGA. Designed 32x32 INT8 systolic array with dataflow optimization executing llama.cpp at silicon level with PS-PL heterogeneous integration via Vitis HLS and Vivado.
- Engineered memory hierarchy with BRAM banking and double-buffering reducing DRAM bandwidth 85%, achieving 120 GOPS throughput at <500mW power for edge inference.
- Demonstrated 6x speedup and 40% efficiency gains over ARM Cortex-A9. Validated via ILA/VIO profiling, establishing FPGA as viable GPU alternative for edge AI.

Pipelined RISC CPU

- Designed 5-stage pipelined RISC processor in Verilog with custom ISA, hazard detection, and forwarding logic. Architected datapath achieving single-cycle throughput for non-dependent instructions.
- Implemented 2-bit dynamic branch predictor achieving 85% accuracy. Reduced pipeline stalls by 60% through speculative execution and branch target buffering.
- Engineered LRU cache hierarchy with 16KB 4-way I-cache and 32KB 8-way D-cache. Implemented write-back and write-allocate policies reducing memory latency 75%.
- Synthesized to 1 GHz on Synopsys 32nm library with <5mm² area. Achieved 100% functional coverage through constrained random verification.

TECHNICAL SKILLS

Languages: Verilog, SystemVerilog, Python, C/C++, Bash, Java

ML/AI Frameworks: PyTorch, TensorFlow, CUDA, NumPy, scikit-learn

Tools: Altera, Questa, Vivado, Vitis, Synopsys DC, Git, Linux, Embedded C, HTCondor